

AI for All

We Need to Talk, AI: a comic essay, carried into the lecture

Prof. Dr.-Ing. Mark Schutera | September 14, 2026 · 7 parts

Seven parts, one comic

Part	Section	The question it answers
I	Intro	why talk about AI at all
II	Basics	what AI is, and what made it work
III	Examples	where it already runs
IV	Chances	what it is good for
V	Risks	what it costs us
VI	Outlook	who gets to shape it
VII	Social Utopias	what we would want it for

cover, the section spiral

How to read this deck

weneedtotalk.ai



One panel caption, **one slide**. Six slides in a row share a heading and a picture: that is one page of the comic, and the six dots under the quote say which panel you are on. Every quote is lifted verbatim. The one italic line under the dots is mine, not theirs.

The comic is a free read at weneedtotalk.ai under CC BY-NC-SA 4.0. It is an essay, not a textbook: it argues. Argue back.

page 5, the authors at the table

[1]

1

Intro

pages 6 to 10

Knowing what future we would like to have

“

Knowing the future is impossible. But what we can do:
Knowing what future we would like to have. And then
work on it.

page 6

page 6, the opening spread

Our creatures have always turned against us

“

Our creatures have always turned against us. This applies to the Golem of Jewish mythology as well as to Frankenstein.

● ● ● ● ● ● page 7

The fear is older than the technology. Hawking only gave it a deadline.

page 7, Golem, Frankenstein and the rubber boots

Our creatures have always turned against us

“

Neither magic nor natural sciences have ever helped to make artificial humans submissive and controllable.

● ● ● ● ● ● ● page 7

The fear is older than the technology. Hawking only gave it a deadline.

page 7, Golem, Frankenstein and the rubber boots

Our creatures have always turned against us

“

Parents whose kids prefer to wear rubber boots instead of weather-proof sandals even when it's hot outside know that.

● ● ● ● ● ● page 7

The fear is older than the technology. Hawking only gave it a deadline.

page 7, Golem, Frankenstein and the rubber boots

Our creatures have always turned against us

“

An AI Take-Over is a hypothetical scenario in which AI takes control of the planet – taking it away from us.

● ● ● ● ● ● page 7

The fear is older than the technology. Hawking only gave it a deadline.

page 7, Golem, Frankenstein and the rubber boots

Our creatures have always turned against us

“

In 2011, Stephen Hawking said that ‘Success in creating AI would be the biggest event in history.’

● ● ● ● ● ● page 7

The fear is older than the technology. Hawking only gave it a deadline.

page 7, Golem, Frankenstein and the rubber boots

Who is Julia

“

Good question. That's something I ask myself, now and then.

● ● ● ● ● ● page 8

Data will not lie to you strategically. It will still lie to you.

page 8, Julia, the digit wall and the 404

Who is Julia

“

I love making sense of our world. I love to discover patterns in data, with code.

● ● ● ● ● ● ● page 8

Data will not lie to you strategically. It will still lie to you.

page 8, Julia, the digit wall and the 404

Who is Julia

“

I love data's friendliness, their lack of strategic answering. If they are biased it's not their fault but ours.

● ● ● ● ● ● ● page 8

Data will not lie to you strategically. It will still lie to you.

page 8, Julia, the digit wall and the 404

Who is Julia

“

Our brains are prone to some errors. Overconfidence. And discrimination against people who seem to be different from us.

● ● ● ● ● page 8

Data will not lie to you strategically. It will still lie to you.

page 8, Julia, the digit wall and the 404

Who is Julia

“

Artificial Intelligence (AI) might have the potential to help us find answers to problems we are not able to solve.

● ● ● ● ● ● page 8

Data will not lie to you strategically. It will still lie to you.

page 8, Julia, the digit wall and the 404

Who is Julia

“

For that reason, we need to demystify AI. Therefore, I'm writing this comic. Let your voice be heard.

● ● ● ● ● ● page 8

Data will not lie to you strategically. It will still lie to you.

page 8, Julia, the digit wall and the 404

Who is Lena

“

Not one thing in particular, I guess.

● ● ● ● ● ● page 9

Ask what the way we draw AI says about what we expect from it.

page 9, Lena, the iceberg of visual codes

Who is Lena

“

I love visual language. I love to encrypt thoughts into pictures by exploiting society's visual memory.

● ● ● ● ● ● ● page 9

Ask what the way we draw AI says about what we expect from it.

page 9, Lena, the iceberg of visual codes

Who is Lena

“

Visuality is one of the most vivid encryption systems I know. We are constantly increasing and renewing it collectively.

● ● ● ● ● ● ● page 9

Ask what the way we draw AI says about what we expect from it.

page 9, Lena, the iceberg of visual codes

Who is Lena

“

It's fun to mess around with visual codes by recomposing associations. Pictures start to shift meanings and interfere in society's processes.

page 9

Ask what the way we draw AI says about what we expect from it.

page 9, Lena, the iceberg of visual codes

Who is Lena

“

AI is a field that gives us an intimate view on society's desires and fears.

● ● ● ● ● ● page 9

Ask what the way we draw AI says about what we expect from it.

page 9, Lena, the iceberg of visual codes

Who is Lena

“

Looking from different angles might help us to overcome commonly known perspectives. Let your view be seen!

page 9

Ask what the way we draw AI says about what we expect from it.

page 9, Lena, the iceberg of visual codes

Lessons from AI in Fiction

“

● ● ● ● ● ● page 10

Every AI in the parade is a character. Not one of them is a system.

page 10, the five fictional robots, 1927 to 2014

Lessons from AI in Fiction

“

● ● ● ● ● ● page 10

Every AI in the parade is a character. Not one of them is a system.

page 10, the five fictional robots, 1927 to 2014

Lessons from AI in Fiction

“

● ● ● ● ● ● page 10

Every AI in the parade is a character. Not one of them is a system.

page 10, the five fictional robots, 1927 to 2014

Lessons from AI in Fiction

“

● ● ● ● ● ● page 10

Every AI in the parade is a character. Not one of them is a system.

page 10, the five fictional robots, 1927 to 2014

Lessons from AI in Fiction

“

● ● ● ● ● ● page 10

Every AI in the parade is a character. Not one of them is a system.

page 10, the five fictional robots, 1927 to 2014

page 10, the five fictional robots, 1927 to 1944

Seems, like we should not seek to replicate but to augment ourselves.

Every AI in the parade is a character. Not one of them is a system.

2

Basics

pages 11 to 19

What is Artificial Intelligence?

“

Actually, the methods are not new. Just the techniques and the data.

● ● ● ● ● ● page 13

The word intelligence is doing work here the technology has not earned.

page 13, the essence of a new situation

What is Artificial Intelligence?

“

At the moment, AI is mainly concerned with performing complicated but repetitive tasks.

● ● ● ● ● ● ● page 13

The word intelligence is doing work here the technology has not earned.

page 13, the essence of a new situation

What is Artificial Intelligence?

“

If you define intelligence as the ability to capture the essence of a new situation...

● ● ● ● ● ● page 13

The word intelligence is doing work here the technology has not earned.

page 13, the essence of a new situation

What is Artificial Intelligence?

“

... real AI needs to think more like us. With intentionality, with emotions, with conscience; capturing counterfactuals.

● ● ● ● ● page 13

The word intelligence is doing work here the technology has not earned.

page 13, the essence of a new situation

What is Artificial Intelligence?

“

This being said, AI is still in its early stages (although it reads differently on the news).

● ● ● ● ● ● page 13

The word intelligence is doing work here the technology has not earned.

page 13, the essence of a new situation

What is Artificial Intelligence?

“

Up to now, it's more adequate to speak of 'artificially augmented intelligence'.*

page 13

The word intelligence is doing work here the technology has not earned.

page 13, the essence of a new situation

Computer Power

“

What kind of techniques helped establish AI? One key indicator is computer power.

● ● ● ● ● ● page 14

Moore said eighteen months. Training compute doubles in three and a half.

page 14, the cost and computer-power curve

Computer Power

“

Moore's law is the observation that the power of computers will double every 18 months.

● ● ● ● ● ● ● page 14

Moore said eighteen months. Training compute doubles in three and a half.

page 14, the cost and computer-power curve

Computer Power

“

The amount of computer power used for training AI has gone up to 3,5 month-doubling time.

● ● ● ● ● page 14

Moore said eighteen months. Training compute doubles in three and a half.

page 14, the cost and computer-power curve

Computer Power

“

The growth has largely been driven by computer hardware, optimized to train and run AI-models.

● ● ● ● ● ● ● ● page 14

Moore said eighteen months. Training compute doubles in three and a half.

page 14, the cost and computer-power curve

Computer Power

“

This explains why AI has been growing faster and faster...

● ● ● ● ● ● page 14

Moore said eighteen months. Training compute doubles in three and a half.

page 14, the cost and computer-power curve

Algorithms 1

“

A second key indicator are better algorithms. Now we can teach machines instead of programming them.

● ● ● ● ● ● page 15

The shift is from writing the rules to supplying the examples.

page 15, a cat? a cat? a dog..?

Algorithms 1

“

Before, when intending to automate a task using AI, we had 1) to understand it completely and 2) to program it.

● ● ● ● ● ● ● page 15

The shift is from writing the rules to supplying the examples.

page 15, a cat? a cat? a dog..?

Algorithms 1

“

The more complicated the task, the harder it was to automate.

● ● ● ● ● page 15

The shift is from writing the rules to supplying the examples.

page 15, a cat? a cat? a dog..?

Algorithms 1

“

The breakthrough was in understanding how we learn to perform tasks.

● ● ● ● ● ● page 15

The shift is from writing the rules to supplying the examples.

page 15, a cat? a cat? a dog..?

Algorithms 1

“

Nowadays, AI learns through trial and error, based on data we feed it. The more data, the more accurate it becomes.

● ● ● ● ● ● page 15

The shift is from writing the rules to supplying the examples.

page 15, a cat? a cat? a dog..?

Algorithms 1

“

Modern AI applications achieve breathtaking results that we could hardly imagine a few years ago.

● ● ● ● ● ● page 15

The shift is from writing the rules to supplying the examples.

page 15, a cat? a cat? a dog..?

Algorithms 2

“

What do these buzzwords even mean? Machine Learning; Deep Learning; Neural Networks (NN)...?

● ● ● ● ● ● page 16

ML, DL and NN are nested, not three names for the same thing.

page 16, the AI bookshelf

Algorithms 2



Machine Learning (ML) helps recognizing patterns based on existing data and algorithms: an important branch of AI.

● ● ● ● ● ● ● page 16

ML, DL and NN are nested, not three names for the same thing.

page 16, the AI bookshelf

Algorithms 2

“

(Note: The AI bookshelf includes other topics, as well.)

● ● ● ● ● page 16

ML, DL and NN are nested, not three names for the same thing.

page 16, the AI bookshelf

Algorithms 2

“

Before Deep Learning (DL), we had to know a lot about the data we were programming. Otherwise, ML would break down.

● ● ● ● ● page 16

ML, DL and NN are nested, not three names for the same thing.

page 16, the AI bookshelf

Algorithms 2

“

DL uses artificial Neural Networks (NN) that learn patterns directly from the data they are fed.

● ● ● ● ● ● page 16

ML, DL and NN are nested, not three names for the same thing.

page 16, the AI bookshelf

Algorithms 2

“

Like our brain, NNs compare new information with objects they know. However, they have not yet been able to think of new objects.

● ● ● ● ● ● page 16

ML, DL and NN are nested, not three names for the same thing.

page 16, the AI bookshelf

Data 1

“

Data is the third element that makes AI so powerful.
But what exactly is data?

● ● ● ● ● ● page 17

Anything you can record is data, which is why nothing is too trivial to collect.

page 17, the data word cloud

Data 1

“

Any form of raw fact or figure is data. Whether on paper or in electronic form.

● ● ● ● ● ● ● page 17

Anything you can record is data, which is why nothing is too trivial to collect.

page 17, the data word cloud

Data 1

“

Data describes what we know and we can draw conclusions from it. Data can take the form of text, numbers, images, or sounds.

● ● ● ● ● page 17

Anything you can record is data, which is why nothing is too trivial to collect.

page 17, the data word cloud

Data 1

“

The internet and mobile devices like smartphones, drones or simple sensors have made data abundant and far more valuable.

● ● ● ● ● page 17

Anything you can record is data, which is why nothing is too trivial to collect.

page 17, the data word cloud

Data 1

“

The more data an algorithm is trained on, the better its results: conclusions, predictions, timing, actions.

● ● ● ● ● ● page 17

Anything you can record is data, which is why nothing is too trivial to collect.

page 17, the data word cloud

Data 1

“

Better results = higher usability = more users = more power. Therefore ‘data is the new oil’.

● ● ● ● ● ● ● page 17

Anything you can record is data, which is why nothing is too trivial to collect.

page 17, the data word cloud

Data 2

“

Okay, all AI applications need training data to get good results.

● ● ● ● ● ● page 18

Four of the five steps are data work. Only the last one is the model.

page 18, the data pyramid, four steps of five

Data 2

“

Only with the invention of the internet, we were able to collect and store enough data to make AI work.

● ● ● ● ● ● page 18

Four of the five steps are data work. Only the last one is the model.

page 18, the data pyramid, four steps of five

Data 2

“

Too much data for us to process. Just the right amount for AI.

● ● ● ● ● page 18

Four of the five steps are data work. Only the last one is the model.

page 18, the data pyramid, four steps of five

Data 2

“

There is an adaption of Maslow's Pyramid of Needs.
Aimed at achieving AI solutions.

● ● ● ● ● page 18

Four of the five steps are data work. Only the last one is the model.

page 18, the data pyramid, four steps of five

Data 2

“

Four out of five necessary steps have to do with data.

● ● ● ● ● ● page 18

Four of the five steps are data work. Only the last one is the model.

page 18, the data pyramid, four steps of five

Data 2

“

AI is a journey that begins with data: collecting, storing, transforming and exploring it.

page 18

Four of the five steps are data work. Only the last one is the model.

page 18, the data pyramid, four steps of five

General AI

“

In the 1960s, AI pioneers hoped that machines could soon learn to think without human intervention.

● ● ● ● ● ● page 19

Everything in the next four parts is narrow AI, not the thing the news means.

page 19, the two AI Winters

General AI

“

But from 1973 onwards, disappointment and criticism in the community, followed by pessimism in the press, led to the 1st ‘AI Winter’. A 2nd one followed.

● ● ● ● ● ● ● page 19

Everything in the next four parts is narrow AI, not the thing the news means.

page 19, the two AI Winters

General AI

“

Until the 2000s, AI was a dirty word that ‘simply didn’t work’. But when data, hardware and infrastructure were ready in 2010, a wide range of AI applications followed.

● ● ● ● ● ● page 19

Everything in the next four parts is narrow AI, not the thing the news means.

page 19, the two AI Winters

General AI

“

Despite their impressive progress and success, today's AI is narrow. Its tasks are often classification and need a lot of data and a lot of energy.

● ● ● ● ● page 19

Everything in the next four parts is narrow AI, not the thing the news means.

page 19, the two AI Winters

General AI

“

No AI can represent causal relationships or integrate abstract knowledge, e.g., what objects are, what they are for, and how they are typically used.

● ● ● ● ● ● page 19

Everything in the next four parts is narrow AI, not the thing the news means.

page 19, the two AI Winters

General AI

“

We really need to adjust our expectations and stop over-hyping AI. If we don't, we may find ourselves in another AI Winter.

page 19

Everything in the next four parts is narrow AI, not the thing the news means.

page 19, the two AI Winters

3

Examples

pages 20 to 27

Maps

“

Using location data from smartphones...

● ● ● ● ● ● page 21

You supplied the training data this morning, on the way in.

page 21, two phones, the delta and the route

Maps

“

... the AI-application Maps can analyze how fast traffic is moving...

● ● ● ● ● ● ● page 21

You supplied the training data this morning, on the way in.

page 21, two phones, the delta and the route

Maps

“

... in real-time.

● ● ● ● ● ● page 21

You supplied the training data this morning, on the way in.

page 21, two phones, the delta and the route

Maps

“

Additionally, maps incorporate user-reported traffic incidents.

● ● ● ● ● ● page 21

You supplied the training data this morning, on the way in.

page 21, two phones, the delta and the route

Maps

“

The proprietary algorithms of Maps have access to vast amounts of data.

● ● ● ● ● ● page 21

You supplied the training data this morning, on the way in.

page 21, two phones, the delta and the route

Maps

“

This means, Maps can suggest the best and fastest routes for us.

page 21

You supplied the training data this morning, on the way in.

page 21, two phones, the delta and the route

Recommendation

“

All major internet companies – or ‘platforms’ – refine their products and processes using AI.

● ● ● ● ● ● page 22

When the platform generates the product too, what is left to choose?

page 22, the component tag cloud

Recommendation

“

If we're looking for books or shoes, internet shops use AI so that our purchase decision gets easier.

● ● ● ● ● ● ● page 22

When the platform generates the product too, what is left to choose?

page 22, the component tag cloud

Recommendation

“

Likewise, streaming providers use AI to analyze our consumer habits down to the last detail.

● ● ● ● ● ● page 22

When the platform generates the product too, what is left to choose?

page 22, the component tag cloud

Recommendation

“

We benefit from improved suggestions and rediscoveries of old pearls which we have missed so far.

● ● ● ● ● ● page 22

When the platform generates the product too, what is left to choose?

page 22, the component tag cloud

Recommendation

“

Also, the platforms use the knowledge of our habits to create and offer new products similar to previous successes...

● ● ● ● ● ● page 22

When the platform generates the product too, what is left to choose?

page 22, the component tag cloud

Recommendation

“

...or to let AI autonomously stir those components together that have worked great elsewhere.

page 22

When the platform generates the product too, what is left to choose?

page 22, the component tag cloud

Social Networks

“

Those of us who use social networks use AI frequently,
without necessarily knowing it.

● ● ● ● ● ● page 23

Invisible by design, which is the precondition for everything else here.

page 23, what a wonderful match

Social Networks

“

Social networks use AI to recommend friends, news, photos, tags, advertisements.

● ● ● ● ● ● ● page 23

Invisible by design, which is the precondition for everything else here.

page 23, what a wonderful match

Social Networks

“

They even ‘curate’ the posts of your friends according to general or personal interests – with the help of AI.

● ● ● ● ● ● page 23

Invisible by design, which is the precondition for everything else here.

page 23, what a wonderful match

Social Networks

“

Now imagine a network that knows even more than our habits, relations, thoughts, interests, likes and dislikes, our hobbies and whereabouts.

● ● ● ● ● page 23

Invisible by design, which is the precondition for everything else here.

page 23, what a wonderful match

Social Networks

“

Our state of health or bank account for example.

● ● ● ● ● ● page 23

Invisible by design, which is the precondition for everything else here.

page 23, what a wonderful match

Social Networks

“

In China, this already exists in the form of 'WeChat'. Just recently, the Chinese government officially announced to supervise each group with 12+ members.

page 23

Invisible by design, which is the precondition for everything else here.

page 23, what a wonderful match

Self-driving Cars

“

Today, we can already activate the autopilot function which automatically and comfortably drives our car or flies our plane.

● ● ● ● ● ● page 24

The blocker is not control. It is explanation.

page 24, the unforeseen circumstance

Self-driving Cars

“

But ultimately, we are responsible. The systems are not autonomous – or self-driving.

● ● ● ● ● ● ● page 24

The blocker is not control. It is explanation.

page 24, the unforeseen circumstance

Self-driving Cars

“

Within a few years, this might change.

● ● ● ● ● ● page 24

The blocker is not control. It is explanation.

page 24, the unforeseen circumstance

Self-driving Cars

“

We would have plenty of room to sleep, work, read or play, sober or drunk after a party.

● ● ● ● ● ● ● page 24

The blocker is not control. It is explanation.

page 24, the unforeseen circumstance

Self-driving Cars

“

Or maybe not. Nearly every car accident involves some sort of unforeseen circumstances, the AI must confront for the first time.

● ● ● ● ● ● page 24

The blocker is not control. It is explanation.

page 24, the unforeseen circumstance

Self-driving Cars

“

Moreover, user tests show that we need autonomous systems to explain their decision to feel safe. Not easy.

page 24

The blocker is not control. It is explanation.

page 24, the unforeseen circumstance

Identifying Images

“

A lot of AI work deals with identifying patterns. At an enormous scale.

● ● ● ● ● ● page 25

If one system reads a tumour and a target, who aims it?

page 25, rifle 2017 next to Picasso 1910

Identifying Images

“

In 2015, computers made fewer mistakes than we do sorting images into predefined categories.

● ● ● ● ● ● ● page 25

If one system reads a tumour and a target, who aims it?

page 25, rifle 2017 next to Picasso 1910

Identifying Images

“

Since 2018, everybody can use a free tool helping AI beginners detect images.

● ● ● ● ● ● page 25

If one system reads a tumour and a target, who aims it?

page 25, rifle 2017 next to Picasso 1910

Identifying Images

“

While early image identification could be lifesaving...

● ● ● ● ● ● page 25

If one system reads a tumour and a target, who aims it?

page 25, rifle 2017 next to Picasso 1910

Identifying Images

“

...sometimes it is life-threatening. The military is a huge fan of it, too.

● ● ● ● ● ● page 25

If one system reads a tumour and a target, who aims it?

page 25, rifle 2017 next to Picasso 1910

Emotional AI

“

Only ~ 10% of the emotional meaning of a message is conveyed through words.

● ● ● ● ● ● page 26

Words are a tenth of the message, so a text interface reads a tenth of you.

page 26, ten per cent of the message

Emotional AI

“

To really understand us, you need our facial expressions, our tone of voice, our gestures plus our words.

● ● ● ● ● ● ● page 26

Words are a tenth of the message, so a text interface reads a tenth of you.

page 26, ten per cent of the message

Emotional AI

“

As our relationship with AI is becoming more and more intimate...

● ● ● ● ● ● page 26

Words are a tenth of the message, so a text interface reads a tenth of you.

page 26, ten per cent of the message

Emotional AI

“

... we need our AI devices (like assistant systems, cars, or robots) to sense and adapt to what we mean and not only to our words.

● ● ● ● ● page 26

Words are a tenth of the message, so a text interface reads a tenth of you.

page 26, ten per cent of the message

Emotional AI

“

Indeed, Emotional AI could help us with some highly valuable tasks. Help autistic children, for example, to learn emotions.

● ● ● ● ● ● page 26

Words are a tenth of the message, so a text interface reads a tenth of you.

page 26, ten per cent of the message

Emotional AI

“

Nonetheless, our freedom of thought might be at stake in hostile scenarios.

page 26

Words are a tenth of the message, so a text interface reads a tenth of you.

page 26, ten per cent of the message

Exoskeletons

“

Exoskeletons are an external framework we can wear to augment our natural physical ability and reduce strain and weaknesses.

● ● ● ● ● ● page 27

A body is a moving target, so the controller has to keep learning.

page 27, elderly man or firefighter

Exoskeletons

“

But every one of us is different.

● ● ● ● ● ● ● page 27

A body is a moving target, so the controller has to keep learning.

page 27, elderly man or firefighter

Exoskeletons

“

We need assistance that is specifically fit for us. We need to tailor our devices at an individual level.

● ● ● ● ● ● page 27

A body is a moving target, so the controller has to keep learning.

page 27, elderly man or firefighter

Exoskeletons

“

AI can use real-time measurements of our body signals, our breathing rate or hip extension for example.

● ● ● ● ● ● page 27

A body is a moving target, so the controller has to keep learning.

page 27, elderly man or firefighter

Exoskeletons

“

This way, the AI can get our individual profile right. AI can make our exoskeletons fit our needs – elderly man or firefighter.

● ● ● ● ● ● ● page 27

A body is a moving target, so the controller has to keep learning.

page 27, elderly man or firefighter

Exoskeletons

“

Our bodies are different and constantly changing. Nowadays, the only feasible way of translating this to robotics is through the help of AI.

● ● ● ● ● ● page 27

A body is a moving target, so the controller has to keep learning.

page 27, elderly man or firefighter

4

Chances

pages 28 to 36

Dealing with Big Data

“

Our power to analyze the tons of data we are now able to store and to collect can be a bottleneck.

● ● ● ● ● ● page 29

The bottleneck was never storage. It is us.

page 29, hundreds of us against the volume

Dealing with Big Data

“

AI helps us deal with large and complex datasets in ways we have never seen before.

● ● ● ● ● ● ● page 29

The bottleneck was never storage. It is us.

page 29, hundreds of us against the volume

Dealing with Big Data

“

Even with hundreds of us analyzing patterns in big data, the sheer volume simply overwhelms our capacities.

● ● ● ● ● ● page 29

The bottleneck was never storage. It is us.

page 29, hundreds of us against the volume

Dealing with Big Data

“

Even if we could analyze data, but it proves to be exhausting and tedious, we can use rule-based AI systems – expert systems – that do the job for us.

● ● ● ● ● page 29

The bottleneck was never storage. It is us.

page 29, hundreds of us against the volume

Dealing with Big Data

“

Traditional algorithms are bound to follow the same logic over and over, without flexibility or learning new paths with new data.

● ● ● ● ● ● page 29

The bottleneck was never storage. It is us.

page 29, hundreds of us against the volume

Efficiency

“

In many situations, AI is simply faster and more accurate than we are. Let's look at an example*.

● ● ● ● ● ● page 30

Ninety-five against eighty-five. What would you need to know to believe it?

page 30, twenty lawyers against one AI

Efficiency

“

A legal platform carried out a competition between 20 experienced lawyers against a trained AI.

● ● ● ● ● page 30

Ninety-five against eighty-five. What would you need to know to believe it?

page 30, twenty lawyers against one AI

Efficiency

“

Competitors had to review 5 legal contracts in 4 hours and identify 30 legal issues.

● ● ● ● ● ● page 30

Ninety-five against eighty-five. What would you need to know to believe it?

page 30, twenty lawyers against one AI

Efficiency

“

They were rated according to how accurately they identified each issue. Who won?

● ● ● ● ● ● page 30

Ninety-five against eighty-five. What would you need to know to believe it?

page 30, twenty lawyers against one AI

Efficiency

“

Humans achieved, on average, an 85% accuracy rate, the AI 95%. Humans took 92 minutes to complete the task, the AI... 26 seconds.

● ● ● ● ● ● page 30

Ninety-five against eighty-five. What would you need to know to believe it?

page 30, twenty lawyers against one AI

Efficiency

“

The human lawyers who competed against the AI in the experiment said, the tasks were very similar to what lawyers do every day.

● ● ● ● ● ● page 30

Ninety-five against eighty-five. What would you need to know to believe it?

page 30, twenty lawyers against one AI

Cognitive Biases

“

Our brain, as a result of evolution, was not optimized for rational decision-making or perfect diagnoses.

● ● ● ● ● ● page 31

A shared distortion does not average out. It compounds.

page 31, gender-bias guiding you through the world

Cognitive Biases

“

Instead, we strive for a competitive degree of ‘fitness’ in our specific environments; quite fast and often satisfying.

● ● ● ● ● ● ● page 31

A shared distortion does not average out. It compounds.

page 31, gender-bias guiding you through the world

Cognitive Biases

“

Our brain power is biologically limited; therefore, we work with heuristics, trust our gut feeling or store & retrieve wrong evidence.

● ● ● ● ● ● page 31

A shared distortion does not average out. It compounds.

page 31, gender-bias guiding you through the world

Cognitive Biases

“

We call this cognitive bias. There are many of them and we share them to some degree.

● ● ● ● ● ● ● ● page 31

A shared distortion does not average out. It compounds.

page 31, gender-bias guiding you through the world

Cognitive Biases

“

And this is what makes our distortions have large-scale consequences. Were they random, they would cancel out.

● ● ● ● ● ● page 31

A shared distortion does not average out. It compounds.

page 31, gender-bias guiding you through the world

page 31, gender-bias guiding you through the world

Used wisely, AI could indeed help us improve our diagnoses and decisions.

A shared distortion does not average out. It compounds.

Creativity

“

AI can save time otherwise wasted on stupid, tedious work for creativity. But it can do even more.

● ● ● ● ● ● page 32

The machine generates. Judging stays with us, and judging is the part we enjoy.

page 32, just a generic style transfer

Creativity

“

When an AI sorts our photos, it is easier to make a photo-calendar with our children (in 28 positions) for our parents as a present.

● ● ● ● ● ● ● page 32

The machine generates. Judging stays with us, and judging is the part we enjoy.

page 32, just a generic style transfer

Creativity

“

When an AI puts together a playlist of up-and-coming producers of house music, it may inspire our next DJ-set.

● ● ● ● ● ● page 32

The machine generates. Judging stays with us, and judging is the part we enjoy.

page 32, just a generic style transfer

Creativity

“

We can also get creative directly with the AI, compiling data and coding it, in a narrow area. But the AI has no judgement.

● ● ● ● ● ● ● page 32

The machine generates. Judging stays with us, and judging is the part we enjoy.

page 32, just a generic style transfer

Creativity

“

Judging is an ability we possess. And we love doing it. This is where AI together with us can achieve great things, with a human tweak.

● ● ● ● ● ● page 32

page 32, just a generic style transfer

Creativity

“

The smaller the role of humans in the process of art-creation becomes, the more AI questions traditional concepts of art.

● ● ● ● ● ● page 32

page 32, just a generic style transfer

Connection

“

Most doctors became doctors because they wanted to connect with people in order to heal them.

● ● ● ● ● ● page 33

The chance is not the diagnosis. It is the time the diagnosis frees up.

page 33, the doctor as overpaid bookkeeper

Connection

“

Today, they often act like overpaid bookkeepers: taking in and spitting back data, prescribing drugs, adjusting doses, ordering tests.

● ● ● ● ● ● ● page 33

The chance is not the diagnosis. It is the time the diagnosis frees up.

page 33, the doctor as overpaid bookkeeper

Connection

“

And it's still only retrospective, error-prone knowledge.

● ● ● ● ● page 33

The chance is not the diagnosis. It is the time the diagnosis frees up.

page 33, the doctor as overpaid bookkeeper

Connection

“

But we cannot connect to doctors who struggle to keep up with messy data from disparate connections. And they cannot connect to us.

● ● ● ● ● page 33

The chance is not the diagnosis. It is the time the diagnosis frees up.

page 33, the doctor as overpaid bookkeeper

Connection

“

AI can support proactive, individualized strategies.

● ● ● ● ● ● page 33

The chance is not the diagnosis. It is the time the diagnosis frees up.

page 33, the doctor as overpaid bookkeeper

Connection

“

Our doctors could get to know us better, having the time to learn how a disease uniquely affects us and educate us in the right way.

page 33

The chance is not the diagnosis. It is the time the diagnosis frees up.

page 33, the doctor as overpaid bookkeeper

New Opportunities

“

AI allows for fascinating new opportunities. Teaching robots without a single line of coding, for example.

● ● ● ● ● ● page 34

Show the robot instead of programming it: page 15's move, applied to a body.

page 34, the sensor jacket and glove

New Opportunities

“

Robots solve complex tasks today – but these tasks are deterministic. At any point of time, you know in advance where your robot needs to be.

● ● ● ● ● ● ● page 34

Show the robot instead of programming it: page 15's move, applied to a body.

page 34, the sensor jacket and glove

New Opportunities

“

Every time a task or location changes, the robot's software must be re-programmed. This is time-consuming, expensive and difficult.

● ● ● ● ● ● page 34

Show the robot instead of programming it: page 15's move, applied to a body.

page 34, the sensor jacket and glove

New Opportunities

“

Even for experienced programmers, it is hard to come up with a set of verbal rules of how to move when movements and settings are tricky or unpredictable.

● ● ● ● ● page 34

page 34, the sensor jacket and glove

New Opportunities

“

AI now enables us to teach robots how to move, either together with the robot, or, with jackets and gloves containing sensors.

● ● ● ● ● ● ● page 34

Show the robot instead of programming it: page 15's move, applied to a body.

page 34, the sensor jacket and glove

New Opportunities

“

From the recorded example-movements, automation scripts can be created and optimized – without us having to write complex code. Fantastic!

page 34

page 34, the sensor jacket and glove

Inclusion

“

AI is also about helping us, about improving inclusion and our well-being, by removing barriers.

● ● ● ● ● ● page 35

Six barriers, one per panel, and not one of them is about convenience.

page 35, a wall missing several bricks

Inclusion

“

AI reads to us, it narrates the world around us and help us ‘see’. Speech-to-text AI helps us ‘hear’, too.

● ● ● ● ● ● page 35

Six barriers, one per panel, and not one of them is about convenience.

page 35, a wall missing several bricks

Inclusion

“

AI helps us type (with words) or talks to us. Users of all ages are adopting virtual assistants because it is so simple to learn how to handle them.

● ● ● ● ● ● page 35

Six barriers, one per panel, and not one of them is about convenience.

page 35, a wall missing several bricks

Inclusion

“

Smart home devices help us do things easily around the house and live independently.

● ● ● ● ● ● page 35

Six barriers, one per panel, and not one of them is about convenience.

page 35, a wall missing several bricks

Inclusion

“

In the classroom, AI can identify individual challenges and offer more personalized approaches.

● ● ● ● ● ● page 35

Six barriers, one per panel, and not one of them is about convenience.

page 35, a wall missing several bricks

Inclusion

“

AI virtual nurses or therapists offer 24-hour-support, or someone to talk to in case we are shy.

● ● ● ● ● ● ● page 35

Six barriers, one per panel, and not one of them is about convenience.

page 35, a wall missing several bricks

Comfort

“

AI satisfies and reinforces our desire for comfort.

● ● ● ● ● ● page 36

What did you stop leaving the house for, and what closed because of it?

page 36, mall_blueprint.pdf

Comfort

“

Almost all answers to our consumer, communication or information needs are just a mouse click or an app away. And most of the time, AI is involved.

● ● ● ● ● ● ● page 36

What did you stop leaving the house for, and what closed because of it?

page 36, mall_blueprint.pdf

Comfort

“

It is tempting not to have to leave our couch at all, no matter what we want to do: eat, work, chat, watch a movie, listen to music...

● ● ● ● ● ● page 36

What did you stop leaving the house for, and what closed because of it?

page 36, mall_blueprint.pdf

Comfort

“

'Pre-AI'-institutions such as shops, local transport, restaurants, libraries, or doctors must react to our new convenience.

● ● ● ● ● page 36

What did you stop leaving the house for, and what closed because of it?

page 36, mall_blueprint.pdf

Comfort

“

All institutions with predominantly analogue offerings are undergoing a tremendous process of change. But it might be worth it.

● ● ● ● ● ● page 36

What did you stop leaving the house for, and what closed because of it?

page 36, mall_blueprint.pdf

Comfort

“

It is precisely these institutions that are social places where we can come together and exchange ideas. Where we can decelerate. 'Waste' time.

page 36

What did you stop leaving the house for, and what closed because of it?

page 36, mall_blueprint.pdf

5

Risks

pages 37 to 43

Surveillance

“

AI makes monitoring people much easier. Not only via our words or facial expressions, via our consumer behavior or interests.

● ● ● ● ● ● page 39

Not only where you are. What you are doing, through a wall.

page 39, the stick figures behind the wall

Surveillance

“

But also, via our movement patterns. Each one of us moves differently. Our movement patterns are unique.

● ● ● ● ● ● ● page 39

Not only where you are. What you are doing, through a wall.

page 39, the stick figures behind the wall

Surveillance

“

Weak radio waves can penetrate walls, but our bodies reflect them. Using this data, an AI can be trained to identify our movements.

● ● ● ● ● ● page 39

Not only where you are. What you are doing, through a wall.

page 39, the stick figures behind the wall

Surveillance

“

That means the AI does not only know where we are.
The AI can also distinguish what we are doing, whether
we are standing, or eating.

● ● ● ● ● ● page 39

Not only where you are. What you are doing, through a wall.

page 39, the stick figures behind the wall

Opinion-Forming and Media

“

AI influences public opinion-forming and the media.
Our public sphere has become fragmented.

● ● ● ● ● ● page 40

Who benefits from the current tools, and how would you find out?

page 40, the fragmented public sphere

Opinion-Forming and Media

“

Borders between private and public communication are blurry. User-generated content is rich but polarized and often privatized on closed platforms.

● ● ● ● ● ● ● page 40

Who benefits from the current tools, and how would you find out?

page 40, the fragmented public sphere

Opinion-Forming and Media

“

New players and sources make special-interest-content popular and visible and, thus, important for mass media.

● ● ● ● ● ● ● page 40

Who benefits from the current tools, and how would you find out?

page 40, the fragmented public sphere

Opinion-Forming and Media

“

Social media-users receive AI-personalized content of political organizations, based on users' personality profiles.

● ● ● ● ● ● page 40

Who benefits from the current tools, and how would you find out?

page 40, the fragmented public sphere

Future Work

“

The share of jobs at high risk of replacement by computers or robots has increased significantly.*

● ● ● ● ● ● page 41

The list is not blue-collar. Lawyers, medics and brokers are on it.

page 41, the job ladder

Future Work

“

This involves all kinds of jobs on the social ladder.
White-collar jobs like lawyers, medics, or brokers...

● ● ● ● ● ● ● page 41

The list is not blue-collar. Lawyers, medics and brokers are on it.

page 41, the job ladder

Future Work

“

... as well as creative jobs like artists, chefs or journalists, or blue-collar jobs such as construction workers or drivers.

● ● ● ● ● ● page 41

The list is not blue-collar. Lawyers, medics and brokers are on it.

page 41, the job ladder

Future Work

“

AI can outperform us in doing repetitive tasks. Cheap, strong, fast, easy.

● ● ● ● ● ● page 41

The list is not blue-collar. Lawyers, medics and brokers are on it.

page 41, the job ladder

Future Work

“

We are better in understanding non-trivial, new situations, building relationships, creating context and meaning. Custom-fit solutions.

● ● ● ● ● ● ● page 41

The list is not blue-collar. Lawyers, medics and brokers are on it.

page 41, the job ladder

Future Work

“

We are also needed as partners of AI: developing algorithms that work, supervising, repairing. Combining us with AI.

page 41

The list is not blue-collar. Lawyers, medics and brokers are on it.

page 41, the job ladder

Data Security and Safety

“

Our blind trust in AI providers is misplaced. We all know that the amount of data that we are implicitly or explicitly forced to leave behind is too great.

● ● ● ● ● ● page 42

Five of the six asks are legislation, not engineering.

page 42, the data letter inbox

Data Security and Safety

“

Since AI needs tons of data, we need to improve our data security and safety. A broad campaign, starting with, e.g.:

● ● ● ● ● ● ● page 42

Five of the six asks are legislation, not engineering.

page 42, the data letter inbox

Data Security and Safety

“

The ‘data letter’, an obligation for all companies and authorities to tell us what data they store about us and how to view, correct and delete it;**

● ● ● ● ● ● page 42

Five of the six asks are legislation, not engineering.

page 42, the data letter inbox

Data Security and Safety

“

Penalties for data outflows for system operators, and an obligation for connected devices to declare the frequency of the security updates (incentives);

● ● ● ● ● ● ● page 42

Five of the six asks are legislation, not engineering.

page 42, the data letter inbox

Data Security and Safety

“

Permanent government investment in secure IT, e.g., open source components written and updated in secure programming languages;

● ● ● ● ● ● page 42

Five of the six asks are legislation, not engineering.

page 42, the data letter inbox

Energy Use

“

In fact, AI has huge potential to optimize energy use, by analyzing complex data, making forecasts, optimizing processes via brain-like algorithms.

● ● ● ● ● ● page 43

The comparison is with a brain, not with last year's data centre.

page 43, save energy, save the planet

Energy Use

“

But conventional hardware was not designed for AI, hence it needs much more computing power, much more energy, than our brains do.

page 43

The comparison is with a brain, not with last year's data centre.

page 43, save energy, save the planet

Energy Use

“

An AI supported translation service, e.g., DeepL, can handle 5,1 quadrillion operations per seconds. But that means masses of computing power and data.

● ● ● ● ● ● page 43

The comparison is with a brain, not with last year's data centre.

page 43, save energy, save the planet

Energy Use

“

Today, data centers use more energy than some countries, e.g., Iran.* In 2018, we had 33 Zettabyte digital data per year. By 2025, we'll have 175.

● ● ● ● ● page 43

The comparison is with a brain, not with last year's data centre.

page 43, save energy, save the planet

Energy Use

“

One of the most worrying forecasts says data centers will use 8% of global energy in 2030. The big internet firms bet on renewable energies.

● ● ● ● ● ● page 43

The comparison is with a brain, not with last year's data centre.

page 43, save energy, save the planet

Energy Use

“

In short, our brains are far more energy efficient than our best computers – and we still do not really understand how.

page 43

The comparison is with a brain, not with last year's data centre.

page 43, save energy, save the planet

6

Outlook

pages 44 to 49

Overall Human-Machine Intelligence

“

We mainly interact with AI via our smartphone. A proprietary marketplace for countless uncured applications (=platform economy).

● ● ● ● ● ● page 45

Your phone analyses you all day. What has it ever helped you think?

page 45, the smartphone as marketplace

Overall Human-Machine Intelligence

“

On the smartphone, we use AI basically for entertainment (image recognition), communication (language and translation) and information (e.g. google).

● ● ● ● ● ● page 45

Your phone analyses you all day. What has it ever helped you think?

page 45, the smartphone as marketplace

Overall Human-Machine Intelligence

“

Unbeknownst to us, AI improves performance and battery life, too. And: It analyzes us, its users.

● ● ● ● ● ● page 45

Your phone analyses you all day. What has it ever helped you think?

page 45, the smartphone as marketplace

Overall Human-Machine Intelligence

“

But couldn't we also develop a smartphone that makes us more creative, more intelligent? That understands like us? Really augment us?

page 45

Your phone analyses you all day. What has it ever helped you think?

page 45, the smartphone as marketplace

Feminism

“

The contribution of women to AI is usually underestimated, especially concerning the democratic applicability of computing.

● ● ● ● ● ● page 46

The first computers were women. The fear is not a takeover but a homogeneous room.

page 46, Ada Lovelace and Grace Hopper

Feminism

“

They were the first computers, actually: running calculations in the first information network of the world.

● ● ● ● ● ● page 46

The first computers were women. The fear is not a takeover but a homogeneous room.

page 46, Ada Lovelace and Grace Hopper

Feminism

“

What we fear is not an AI-takeover. We fear scientists' homogeneity leading them to (finally!) 'create' intelligent beings – in an egocentric way.

● ● ● ● ● ● page 46

The first computers were women. The fear is not a takeover but a homogeneous room.

page 46, Ada Lovelace and Grace Hopper

Feminism

“

In an AI world, where ‘female’ service robots are ‘hot’ and submissive, how can children learn to treat women with respect, with dignity?

● ● ● ● ● ● ● page 46

The first computers were women. The fear is not a takeover but a homogeneous room.

page 46, Ada Lovelace and Grace Hopper

Feminism

“

And many of those whose service jobs will be replaced belong to an already vulnerable group: minority women.

● ● ● ● ● ● page 46

The first computers were women. The fear is not a takeover but a homogeneous room.

page 46, Ada Lovelace and Grace Hopper

Feminism

“

We need (different!) women to participate in the development and use of AI. Otherwise we'll just reproduce patriarchal structures.

page 46

page 46, Ada Lovelace and Grace Hopper

Digital Capitalism

“

Algorithms have always been an integral part of our modern capitalist societies.

● ● ● ● ● ● page 47

The paradigm is not production and sale. It is collection and use.

page 47, Weber's judge and Marx's worker

Digital Capitalism

“

The idea of objectiveness, general abstract rules, quantification, de-personalization and the concept of equality are characteristics of modernity.

● ● ● ● ● ● ● page 47

The paradigm is not production and sale. It is collection and use.

page 47, Weber's judge and Marx's worker

Digital Capitalism

“

Max Weber, e.g., pictures the judge as an automation who processes concrete cases (data) based on an algorithm, namely the law.

● ● ● ● ● ● ● page 47

The paradigm is not production and sale. It is collection and use.

page 47, Weber's judge and Marx's worker

Digital Capitalism

“

Marx' workers are an appendage of the machine, alienated from the spiritual potencies of the capitalist working process.

page 47

The paradigm is not production and sale. It is collection and use.

page 47, Weber's judge and Marx's worker

Digital Capitalism

“

Digital capitalism is the ground on which AI thrives. The central paradigm is – instead of production and sale – the collection and use of data.

page 47

The paradigm is not production and sale. It is collection and use.

page 47, Weber's judge and Marx's worker

Digital Capitalism

“

Since today's AI is largely incomprehensible to us, we need to regulate and shape AI as a public service and snatch the omnipotence over AI from the few dominant, digital companies.

page 47

The paradigm is not production and sale. It is collection and use.

page 47, Weber's judge and Marx's worker

ART: Accountability, Responsibility, Transparency

“

Throughout the world, we share a growing awareness:
We need to make AI safe, beneficial and fair for us.

● ● ● ● ● ● page 48

The acronym is the easy part. The governance it asks for does not exist.

page 48, the three equal pillars

ART: Accountability, Responsibility, Transparen

“

This requires the participation and commitment from many of us. It means training, regulation and awareness in 3 equal pillars:

● ● ● ● ● ● ● page 48

The acronym is the easy part. The governance it asks for does not exist.

page 48, the three equal pillars

ART: Accountability, Responsibility, Transparency

“

Accountability is the concept that AI should be held responsible for the results of its algorithms.

● ● ● ● ● ● page 48

The acronym is the easy part. The governance it asks for does not exist.

page 48, the three equal pillars

ART: Accountability, Responsibility, Transparen

“

This includes our responsibility to develop a framework for AI that incorporates our values.

● ● ● ● ● ● page 48

The acronym is the easy part. The governance it asks for does not exist.

page 48, the three equal pillars

ART: Accountability, Responsibility, Transparen

“

Transparency refers to the need to describe, inspect and reproduce the AI algorithms and results, and to manage the data used, in a fair way.

● ● ● ● ● ● page 48

The acronym is the easy part. The governance it asks for does not exist.

page 48, the three equal pillars

No Fear of Numbers

“

In order to understand how AI could improve our life,
we propose a real revolution:

● ● ● ● ● ● page 49

Who in this room went quiet when the first equation appeared?

page 49, 1908 against MCMVIII

No Fear of Numbers

“

Healing the public shaming in our individual
math-biographies that caused fear.

● ● ● ● ● ● ● page 49

Who in this room went quiet when the first equation appeared?

page 49, 1908 against MCMVIII

No Fear of Numbers

“

Many of our brains signal ‘pain’ when we read numbers, think or talk about them.

● ● ● ● ● ● ● page 49

Who in this room went quiet when the first equation appeared?

page 49, 1908 against MCMVIII

No Fear of Numbers

“

Not only does this deprive life. Zero, e.g., is a fascinating tool allowing us to write '1908' instead of MCMVIII: $1,000 + (1,000 - 100) + 5 + 1 + 1 + 1$.

page 49

Who in this room went quiet when the first equation appeared?

page 49, 1908 against MCMVIII

No Fear of Numbers

“

Zero is also part of the binary code, fundamental to today's computers and their language. Fundamental to AI, therefore.

● ● ● ● ● ● ● page 49

Who in this room went quiet when the first equation appeared?

page 49, 1908 against MCMVIII

No Fear of Numbers

“

With improved 'math-esteem', more of us could and should discuss chances and risks of AI in our lives.

page 49

Who in this room went quiet when the first equation appeared?

page 49, 1908 against MCMVIII

7

Social Utopias

pages 50 to 53

AI today is not Skynet

“

For over 300,000 years we've been hunting with spears;
for less than ten years we've been looking for our next
song with the help of AI.

page 51

page 50, the section opener

It depends on the training data

“

If we train AIs on discriminatory data, they will make discriminatory decisions and reduce, not increase, the opportunities, and resources of the underprivileged.

page 52

page 52, the application procedure

Raising the question, not settling it

“

An AI may not be able to resolve all injustices, but it can raise questions about how we want to live.

page 52

page 52, the motorway through the city

Or a creepy surveillance state

“

Or is this vision making us create a creepy surveillance state? Why should the authorities always be well-meaning? How could we make sure they are?

page 53

page 53, the same platform, read twice

What do you think?

“

But the first step has been made: What can AI help us with? Where do we have to be careful? What do you think? Let your voice be heard and your point of view be seen.

page 53

page 53, Tbc.

Further reading, the book's own

Authors	Title
Agrawal, Gans, Goldfarb	Prediction Machines: The Simple Economics of Artificial Intelligence, 2018
Evans	Broad Band: The Untold Story of the Women Who Made the Internet, 2018
Chollet, Allaire	Deep Learning with R, 2018
Lee	AI Superpowers: China, Silicon Valley, and the New World Order, 2018
Bostrom	Superintelligence: Paths, Dangers, Strategies, 2014
Zuboff	The Age of Surveillance Capitalism, 2018
Russell, Norvig	Artificial Intelligence: A Modern Approach, 2010
Daum	Das Kapital sind wir: Zur Kritik der digitalen Ökonomie, 2017

page 55, the reading list

[1]

If you want to build one

the track



This deck is about what AI does to us. For how it is built, work Google's **foundational courses** track, four courses in order:

Course	Length	Answers
Introduction to Machine Learning	20 min	what ML is
Machine Learning Crash Course	12 modules	how to build one
Problem Framing	45 min	whether to at all
Managing ML Projects	90 min	how to keep it alive

developers.google.com/machine-learning/foundational-courses

a laptop on the same table

[2]

References

- 1 Julia Schneider and Lena Kadriye Ziyal. *We Need to Talk, AI: A Comic Essay on Artificial Intelligence*. Berlin: Dr. Julia Schneider, 2019. ISBN: 978-3-748531-28-9. Free read, CC BY-NC-SA 4.0. <https://weneedtotalk.ai>.
- 2 Google Developers. *Foundational Courses: Machine Learning*. Course hub. <https://developers.google.com/machine-learning/foundational-courses>. 2025.
- 3 Google Developers. *Introduction to Machine Learning*. <https://developers.google.com/machine-learning/intro-to-ml>. 2025.
- 4 Google Developers. *Machine Learning Crash Course*. <https://developers.google.com/machine-learning/crash-course>. 2025.
- 5 Google Developers. *Introduction to Machine Learning Problem Framing*. <https://developers.google.com/machine-learning/problem-framing>. 2025.
- 6 Google Developers. *Managing Machine Learning Projects*. <https://developers.google.com/machine-learning/managing-ml-projects>. 2025.

**Let your voice be heard
and your point of view be seen**

Prof. Dr.-Ing. Mark Schutera | DHBW Ravensburg